

Scientific Computing

Mathematical Foundations for Machine Learning

Seminar 3: Problem Set

Mixtures, Dependence, and Random Vectors

Block I, following Lectures 2 and 3

Problem 1 (Inverted dropout). During training, inverted dropout replaces an activation x by

$$y = \frac{mx}{1-p}, \quad m \sim \text{Bernoulli}(1-p),$$

where m is independent of x ; at test time $y = x$. Let $\mathbb{E}[x] = \mu$ and $\text{Var}(x) = s^2$. Find $\mathbb{E}[y]$ and $\text{Var}(y)$. How many times larger is the variance of the activation during training than at test time for $p = 0.5$, $\mu = 1$, $s^2 = 1$?

Problem 2 (Uniform distribution on a disk). The point (X, Y) is uniformly distributed on the unit disk $\{x^2 + y^2 \leq 1\}$. Find the marginal density of X , the covariance $\text{Cov}(X, Y)$, the conditional mean $\mathbb{E}[Y | X = x]$, and the conditional variance $\text{Var}(Y | X = x)$. Are X and Y independent? What is the best predictor of Y from X in the mean-squared sense?

Problem 3 (Attention with correlated features). Let $q, k \in \mathbb{R}^d$ be independent with $q, k \sim \mathcal{N}_d(0, \Sigma)$. Show that $\text{Var}(\langle q, k \rangle) = \text{tr}(\Sigma^2)$. For $\Sigma = (1 - \rho)I_d + \rho \mathbf{1}\mathbf{1}^\top$ with $0 \leq \rho < 1$, find the variance of the scaled attention logit $z = \langle q, k \rangle / \sqrt{d}$. What happens to $\text{Var}(z)$ as $d \rightarrow \infty$ for fixed $\rho > 0$?

Hint: condition on k .

Problem 4 (The reparameterisation trick). A variational autoencoder samples a latent vector $z \sim \mathcal{N}_2(\mu, \Sigma)$ as $z = \mu + A\varepsilon$ with $\varepsilon \sim \mathcal{N}_2(0, I)$, so that gradients can flow through μ and A . Let

$$\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 5 \end{pmatrix}.$$

Find a lower-triangular matrix A for which $z \sim \mathcal{N}_2(\mu, \Sigma)$. Is it the only matrix A with this property? Find the distribution of $z_1 - z_2$.

Problem 5 (Is the naive Bayes assumption violated?). A class $C \in \{-1, +1\}$ takes both values with probability $\frac{1}{2}$. Given C , two features are independent, with

$$X_i | C \sim \mathcal{N}(\mu C, 1), \quad i = 1, 2, \quad \mu > 0,$$

which is exactly the assumption of a naive Bayes classifier. A colleague computes the correlation between X_1 and X_2 over the whole dataset, obtains 0.5, and concludes that the naive Bayes assumption does not hold. Find $\text{Corr}(X_1, X_2)$ as a function of μ . Is the colleague right?

Bonus problems

Problem 6* (The best of two seeds). Two training runs of the same model with different random seeds have standardised validation scores X and Y , which are jointly Gaussian with zero means, unit variances, and correlation ρ . You keep the better run. Find $\mathbb{E}[\max(X, Y)]$. How much does the second seed give you for $\rho = 0$ and for $\rho = 0.9$?

Hint: $\max(a, b) = \frac{a+b}{2} + \frac{|a-b|}{2}$, and $\mathbb{E}|Z| = \sqrt{2/\pi}$ for $Z \sim \mathcal{N}(0, 1)$.

Problem 7* (The optimal denoiser for Gaussian data). A diffusion model corrupts a clean sample x_0 as

$$x_t = \sqrt{\bar{\alpha}} x_0 + \sqrt{1 - \bar{\alpha}} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1), \quad 0 < \bar{\alpha} < 1,$$

with ε independent of x_0 , and a network is trained with the mean squared error to predict x_0 from x_t . Suppose the data are $x_0 \sim \mathcal{N}(0, 1)$. Find the best possible prediction $\mathbb{E}[x_0 | x_t]$ and the remaining uncertainty $\text{Var}(x_0 | x_t)$. What does the optimal denoiser output when $\bar{\alpha}$ is close to 0 and when it is close to 1?